

Workforce pathways into the semiconductor manufacturing sector: Mechanisms and Interventions

Specific Aims

Aim 1: Advance workforce pathways into the semiconductor manufacturing sector via a field experiment-focused strategy.

Aim 2: Advance workforce pathways into the semiconductor manufacturing sector via a field experiment-focused strategy.

Aim 3: Advance workforce pathways into the semiconductor manufacturing sector via a field experiment-focused strategy.

Background

Our preliminary data suggest a tractable path toward measurable improvement. We will assemble a multidisciplinary team with complementary expertise. This proposal addresses workforce pathways into the semiconductor manufacturing sector, a problem with substantial significance. The approach integrates established field experiment methods with a novel analytical pipeline.

Approach

We will assemble a multidisciplinary team with complementary expertise. The approach integrates established field experiment methods with a novel analytical pipeline. Findings will be disseminated through peer-reviewed publication and open data release. Prior work has established a partial understanding, yet key gaps remain.

Innovation

Prior work has established a partial understanding, yet key gaps remain.

Investigator Context

The PI's first independent award; strong mentorship is in place. The R1 host institution offers extensive core facilities and shared instrumentation, backed by a deep institutional record of federally funded work.

Budget Justification

Total requested support is \$1,611,680 over 4 years at a R1 institution, covering personnel, materials, and dissemination.

References

1. [1] <https://doi.org/10.1371/journal.pone.0173664>
2. [2] <https://doi.org/10.1126/science.1259855>
3. [3] <https://doi.org/10.1056/NEJMoa2034577>
4. [4] <https://doi.org/10.1073/pnas.1517384113>